Network Manager Release Notes

Version 2.0

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1: Introduction

This document provides a summary of the major changes in this release of Network Manager for Telecoms.

1.1 Intended Audience

This document is intended for Administrators and Developers with a detailed technical knowledge of IQGeo Platform and utilities.

2: Installation

The following changes relate to installation and upgrade.

2.1 Prerequisites

Component	Version	Comments
IQGeo Platform	6.1	Mandatory
IQGeo Workflow Module	6.1	Optional
IQGeo Anywhere	6.1.3 (or higher)	Optional

All components must include any released patches.

2.2 Installing the Software

1. Uncompress the software download and copy the **comms** folder into:

   ```
   <myWorld_install_dir>/WebApps/myworldapp/modules
   ```

2. Download the module's JavaScript dependencies:

   ```
   myw_product fetch node_modules
   ```

3. Rebuild your JavaScript bundles:

   ```
   myw_product build applications
   myw_product build config
   ```

2.3 Creating a New Database

If you do not already have an IQGeo database, or that database does not yet have the Network Manager tables installed, you will need to:

- Install the Network Manager tables.
- Configure the network object types for your installation.

These processes are covered in detail in the *Installation and Configuration Guide*.

2.4 Upgrading an Existing Database

If you already have a Network Manager database from a previous release, you will need to:

- Upgrade your database.
- Perhaps perform some post-upgrade configuration changes.

These processes are covered in detail in *Upgrading* on page 22.

3: Functional Changes

3.1 Schematic View Added

Schematics provide a graphical representation of your data. They are generated on-the-fly and updated automatically as your work. You can pan and zoom a schematic and select objects in a similar way to the map.

Two types of schematic are supported at this release:

- The *Structure Connectivity Schematic* shows equipment and connections at a structure.

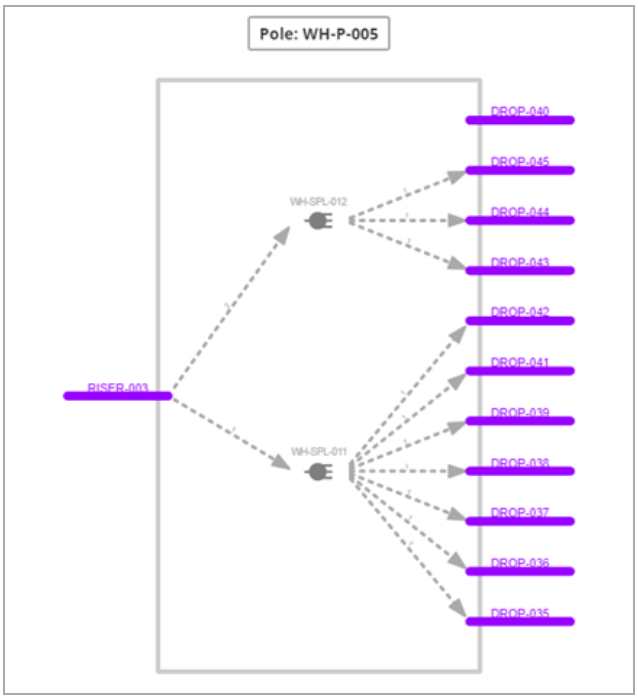

Figure 3-1: Structure Connectivity Schematic

- The *Fiber Trace Schematic* shows graphical view of a fiber trace.

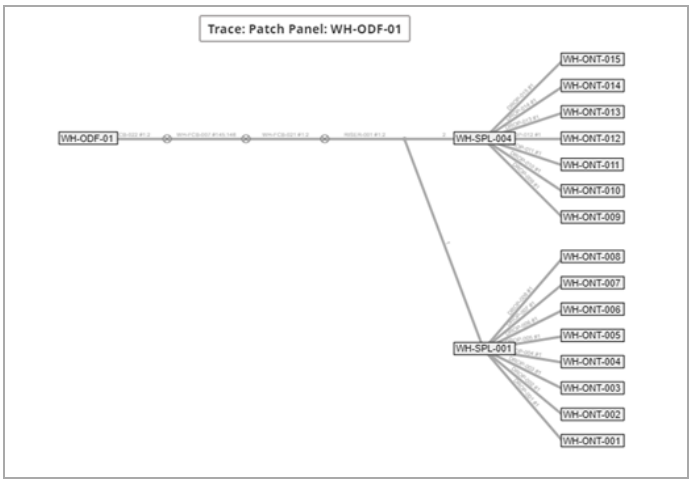

Figure 3-2: Fiber Trace Schematic

It is anticipated that additional types of schematic will be added in future releases. It is also possible to add custom types if desired.

For more details, see the *User Guide*.

3.2 Inside Plant Support Added

The system has been extended to support detailed modeling of network equipment at customer premises and hubs.

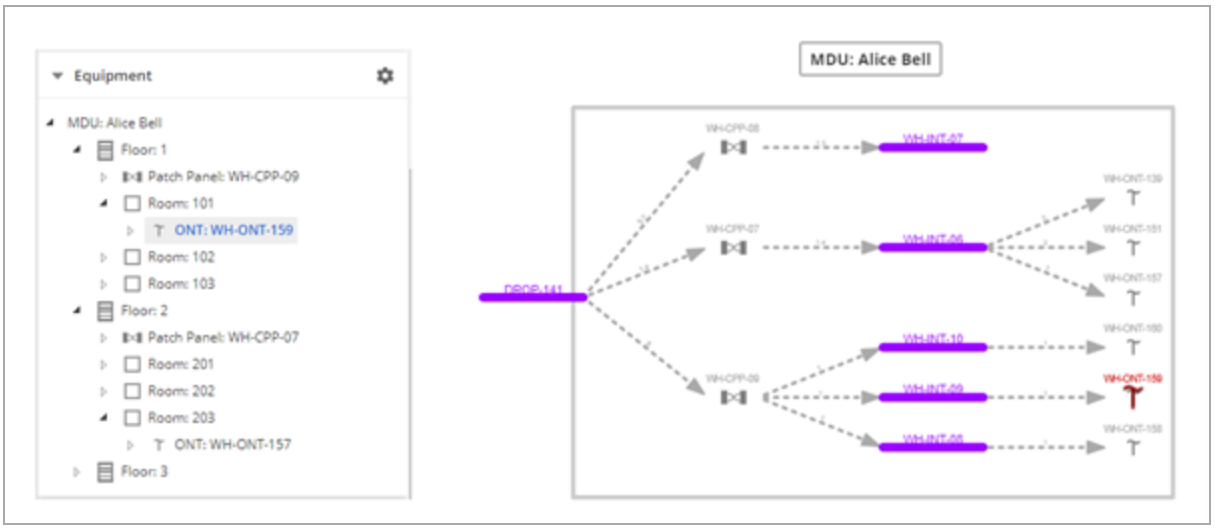

Figure 3-3: Structure Schematic (Inside Plant)

Changes to support internal plant include:

- New objects MDU, floor and room object in the fiber template model.
- Ability to add internal cables within a structure.
- Support for direct port-to-port connections.

These functions, together with *Structure Connectivity* schematic, allow users to build a detailed model of containment and connectivity within a structure.

3.3 Cable Slack Support Enhanced

It is now possible to add multiple slack loops on a cable within a given structure. This allows more accurate modeling of fiber network paths.

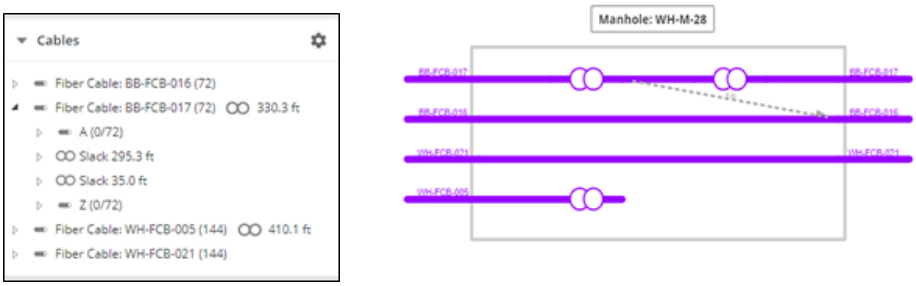

Figure 3-4: Cable Slack Schematic

To support the new functionality, the **Cable Tree Add Slack** option has moved from cable node to its **IN**/**OUT** nodes. There is also a new option to split an existing slack.

3.4 Logical Inventory Support Added

Network Manager now allows you to record the paths used by network services (*circuits*). This permits designers to see at a glance which equipment is available when provisioning new services.

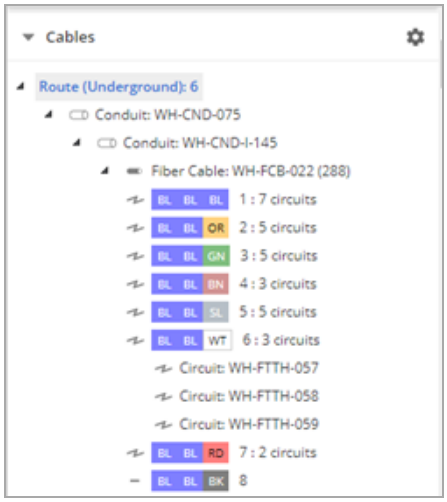

Figure 3-5: Circuits

The equipment and cable trees now show which ports and fibers have circuits running on them.

It is also possible to view circuits that have been routed within designs but not yet published (see *Conflict Avoidance Added* on page 13).

Circuits are routed using a specialized editor and associated routing dialog.

Figure 3-6: Circuit Editor and Routing Dialog

The system automatically prevents disconnections and other network edits that would break a circuit path.

Circuit support is enabled using the **Settings** configuration page. For more information, see the *Installation and Configuration Guide*.

3.5 Design Rule Support Added

There is a new framework for defining custom design rules, for example to check pole spacing or conduit capacity constraints. Design rules are applied automatically as part of the design check.

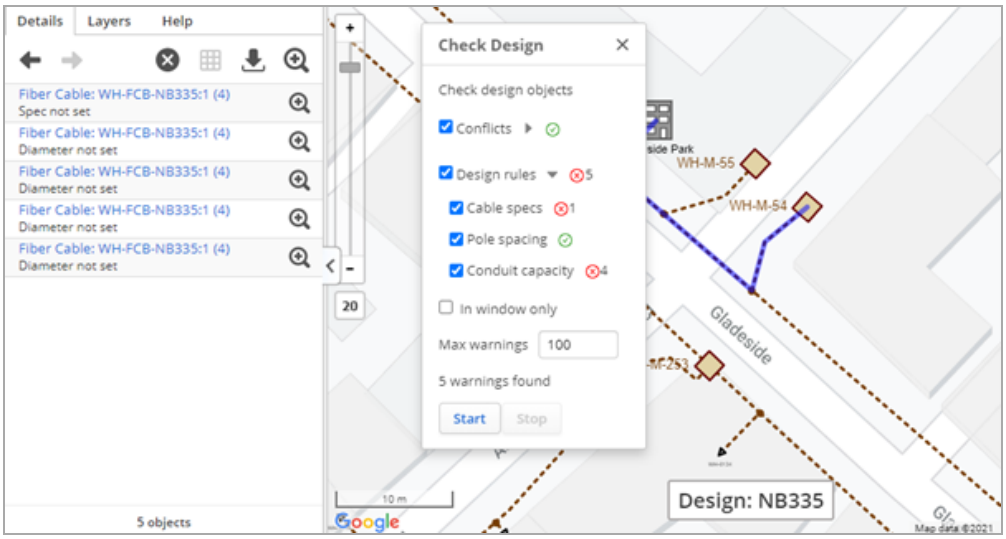

Figure 3-7: Design Rules

Design rules are coded in JavaScript and registered via your application's JavaScript definition file. For more details, contact IQGeo Support.

3.6 Conflict Avoidance Added

The **Equipment** and **Cable** trees can now show objects from other designs, for example planned cables and connections. This helps avoid conflicts later.

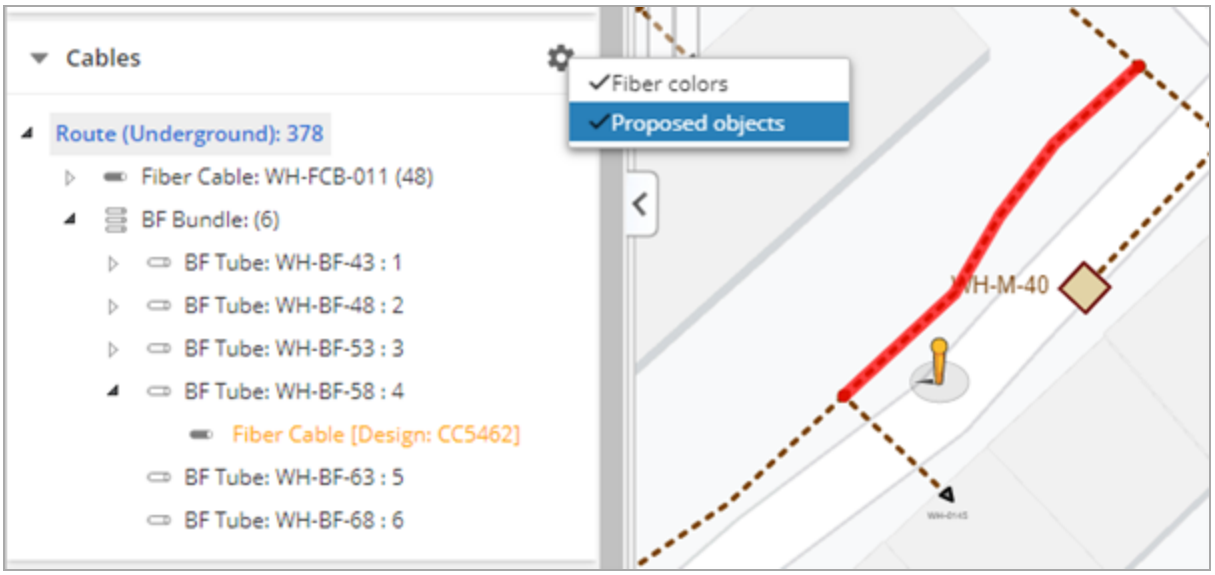

Figure 3-8: Proposed Objects

Proposed objects are shown in orange. They include a link to the design in which they were created. Clicking on the link will take you to the design (if it accessible to you).

Conflict avoidance is supported for conduits, cables, equipment, connections and circuits. Proposed structures and routes can already be visualized using *forward view* map layers.

3.7 Conflict Resolution Enhanced

The display of conflict information in the GUI has been enhanced to include more information.

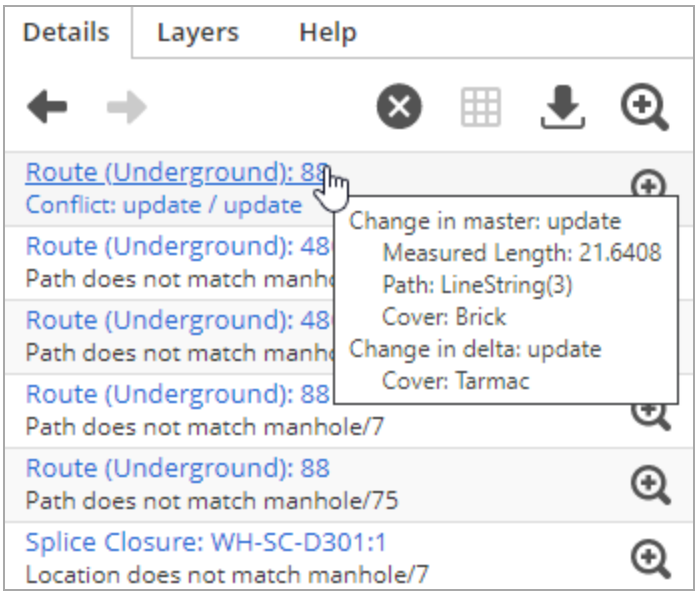

Figure 3-9: Conflict Resolution

In addition, auto-resolution of conflicts via the **Fix Conflicts** function now handles more cases:

- Geometry and derived attribute conflicts are now auto-resolved on all contained object types.
- Update conflicts are auto-merged where attributes changes are identical.
- Delete conflicts are auto-merged where the conflicting change is to derived fields only.

For more information, see the *User Guide*.

3.8 Offline Use Now Supported

Network Manager can now be used in conjunction with IQGeo Anywhere to view and update network data whilst working offline. The offline system supports both master data and designs. All standard Network Manager functions are available offline except conflict fix-up and design publishing.

For information about enable offline working, see the *Installation and Configuration Guide*.

4: End-user GUI Changes

4.1 Toolbar

The main tool bar has been reorganized as follows:

- There is a new item **Schematics View**.
- The **Spec Manager** and **Data Validation** buttons have been moved to the **Tools** palette.
- The **Tools Palette** button is now displayed before the **Structure Palette** button.

These changes are intended to reduce the clutter in the main toolbar. For more details, see the *User Guide*.

4.2 Equipment Tree

The **Equipment Tree** has been modified as follows:

- Circuits: Port nodes now indicate if the port has circuits running on it (see *Logical Inventory Support Added* on page 10).
- Show usage: Context menu item **Show Usage** has been renamed as **Show Terminations**.
- Show circuits: There is a new context menu option **Show Circuits**. This retrieves detailed information about the circuits running on a port (or set of ports).
- Internal cables: The context menu for structure nodes now supports adding an internal cable. This can be useful when modeling MDUs and equipment rooms (see *Inside Plant Support Added* on page 9).
- Options: There is a new option **Proposed Objects** that controls the display of objects from other designs (see *Conflict Avoidance Added* on page 13).
- Port connection: The **Port Connection** dialog now supports direct port-to-port connections. This can be useful when modeling MDUs and equipment rooms (see *Inside Plant Support Added* on page 9).

For more information, see the *User Guide*.

4.3 Cable Tree

The Cable Tree has been modified as follows:

- Slacks: The **Add Slack** function has been moved from cable nodes to segment nodes. This gives more control over where the slack is added relative to any existing connections. Slack objects show as separate nodes with the tree and support a new function **Split** (see *Cable Slack Support Enhanced* on page 10).

- Internal segments: Cable nodes now include a sub-node for each internal cable segment within a structure. This permits examination and modification of internal connectivity.
- Circuits: Fiber nodes now indicate if the fiber has circuits running on it (see *Logical Inventory Support Added* on page 10).
- Show circuits: There is a new context menu option **Show Circuits**. This retrieves detailed information about the circuits running on a fiber (or set of fibers).
- Show usage: Context menu item **Show Usage** has been renamed **Show Terminations**.
- Options: There is a new option **Proposed Objects** that controls the display of objects from other designs (see *Conflict Avoidance Added* on page 13).

4.4 Digitizing

When inserting a route object, the system now splits the route at each intermediate structure (rather than just connecting at end points). This speeds up the process of creating a route linking a set of structures.

Splitting is also applied during route update.

4.5 Palettes

The structure and equipment palettes not longer default to **autoSave** mode. Changes are now only saved when the user clicks the **Save** button.

If desired, **autoSave** mode can be re-enabled by modifying the parameters of **StructureModePlugin** and **EquipmentModePlugin** in your application's JavaScript definition file.

4.6 Design Toolbar

The Design Toolbar has been modified as follows:

- Design changes: **Select objects** now provides more details on the changes made in a design. To see which fields have been changed on an object, hover over its entry in the results list.
- Conflicts: The summary information displayed for conflicts and integrity errors in the **Details** pane has been improved. In addition, you can view changes at attribute level by hovering over entries in the results list.
- Fixup: You can now see which fields of an object were modified by the system during automated conflict resolution by hovering over its entry in **Fix conflict** results.
- Check design dialog: There is a new **Check Design** dialog that provides feedback on validation progress. It also includes options for limiting the check to specific categories and to the window bounds.
- Design rules: The **Check Design** dialog can also be used to run design rules, if configured.

As noted above, automated conflict resolution is enhanced at this release to cover more cases. In addition, it is now possible to resolve conflicts manually by changing record attributes to match master values.

For more information, see the *User Guide*.

5: Administrator GUI Changes

5.1 Data Validation Tool

The following changes have been made to the Data Validation tool:

- Activation: The tool is now activated from **Tools** palette (rather than from the main toolbar).
- Design rules: The tool now supports checking design rules (if configured).
- GUI: The dialog now provides feedback on validation progress and provides an interrupt button.
- Checks: Validation has been extended to include checks for duplicate and out-of-range connections.

Access to **Data Validation** tool is now controlled by a separate right (see *Roles Configuration Page* on page 18).

5.2 Specifications Manager

The following changes have been made to the **Specification Manager** tool:

- Activation: The tool is now activated from **Tools Palette** (rather than from the main toolbar).
- Editing: It is now possible to edit properties of a specification.

Note that edits to specification records should normally be confined to meta-data fields only. Changes to physical fields are not propagated to existing objects that use the specification.

5.3 Settings Configuration Page

The **Comms Settings** page has been modified as follows:

- Circuits tab: There is a new tabs **Circuits** for configuring circuit feature types.
- Designs tab: There is a new tab **Designs** for configuring delta owner feature types. This replaces JavaScript configuration in previous releases.
- Other: Existing tabs have been reordered and the **Cable Preview Styles** tab has been renamed as **Styles**.

For more information, see the *Installation and Configuration Guide*.

5.4 Roles Configuration Page

There is a new Comms right **Run data validation** that controls access to the **Data Validation** tool. You may need to grant this right to existing users (see *Upgrading* on page 22).

6: Data Model Changes

6.1 Fiber Template Model Modified

The fiber template model has been modified as follows:

- Drop Point: There is a new structure type **drop_point** that models a fiber customer access point (flowerpot or Toby).
- MDU: There is a new structure type **mdu** that models a multi-dwelling unit.
- Floor and Room: There are new equipment types **floor** and **room** that model containment within a building or mdu.
- FTTH Circuit: There is a new feature type **ftth_circuit** that models a fiber-to-the-home circuit.
- Patch Panel: The function of equipment type **fiber_patch_panel** has been changed from **cross-connect** to **connector**. This permits tracing through patch panels.

Note that these changes affect new installations of the template models only. Existing models are not modified by the upgrade.

6.2 Conduit Data Model Modified

The system data model is modified as follows:

- Blown Fiber Tubes: The elements of a conduit run no longer need to follow the same direction. This avoids the need to reverse conduits when joining tubes (which increased the risk of conduits).

  Existing reversed conduits are fixed up by the database upgrade. The **forward** flag on conduit objects will be removed in a future release.

This may affect your data import scripts.

7: API Changes

7.1 Design Configuration

The list of feature types that can own a delta is now configured via a database setting rather than in code. It is no longer necessary to set the **deltaOwners** property on **CommsDeltaOwnerPlugin** when enabling versioning (and any existing configuration should be removed).

7.2 Server-side Triggers

There is a new server-side mechanism for firing actions when a network object is updated.

For details, see the *Developer Guide*.

7.3 Name Manager

Code specific to name managers is removed at this release. The setting **mywcom.nameManager** is no longer used and is dropped by the database upgrade.

If you have implemented a custom name manager, you will need to convert it to use the server-side trigger mechanism.

This change is to support the use of custom name managers when working offline.

7.4 Fiber Color Manager

JavaScript class **FiberColorManagerPlugin** is renamed to **DisplayManager** at this release.

This change is to better reflect the responsibilities of the class.

7.5 Custom Tools

The signature required for plugin buttons added to the **Tools Palette** has changed as follows:

- **Method active()** is renamed to **isActive()**. This change is to avoid clash with core **.active** property.
- There is a new method **hasPermission()** that controls if the button is shown. This allows the palette contents to be changed dynamically based on user permissions and design state.

8: Documentation Changes

The following documentation changes have been made at this release:

- User Guide: The *User Guide* has been organized and extended.
- Installation and Configuration Guide: The *Installation and Configuration Guide* now includes step-by-step instructions for adding a new network object type.
- Developer Guide: There is a new guide that provides detailed information, including a description of the system architecture and data model, for developers.

The documentation is published in a responsive HTML format that adapts to the user's device and facilitates searching across guides.

9: Upgrading

This section provides an overview of the upgrade process for this release.

9.1 Upgrading Your Database

To upgrade from the previous release of Network Manager:

1. Apply all IQGeo Platform upgrades:

   ```
   myw_db <db_name> upgrade core 600 610
   ```

2. Apply the Network Manager upgrade:

   ```
   myw_db <db_name> upgrade comms 6120
   ```

3. Perform the post upgrade actions listed below.

9.2 Post-upgrade Actions

After upgrading your database, you should:

1. Edit your application's JavaScript definition to remove any configuration of **deltaOwners** (see *API Changes* on page 20).
2. Using the **Comms Settings** configuration page, register you delta owners in database setting **mywcom.designs**.
3. Using the **Roles** configuration page, grant right **Comms: Run data validation** to your administrator and any other roles that require it.

You should also review any custom code for impact from the changes described in *API Changes* on page 20.

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